

Amar Telidji University Laghouat

Department: Computer Science

Level: 4th Year Engineer

Course: Machine learning, Deep learning and Security

Assignment CH03 Report: Malware Classification
using MLP

Dataset: Malware Analysis Dataset - API Call
Sequences (Kaggle)

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1. Introduction

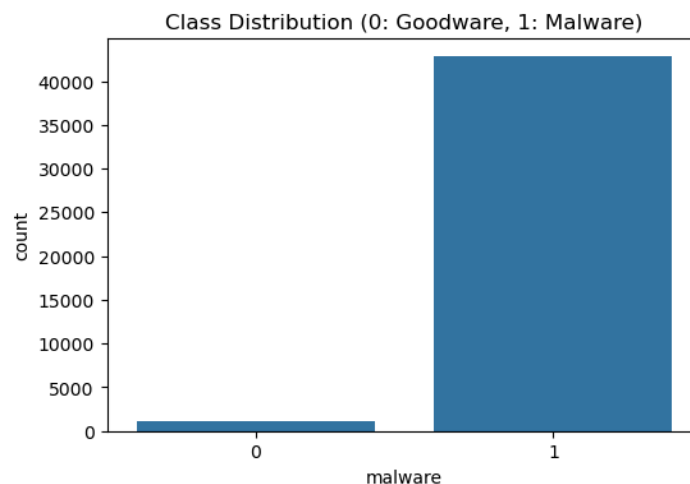
The objective of this assignment is to explore the capabilities of Deep Learning, specifically Multi-Layer Perceptrons (MLP), in cybersecurity. We aim to classify software samples as **Malware (1)** or **Goodware (0)** using sequential API call data. This report details the experimental setup, analyzes the performance of different neural network architectures, and compares these Deep Learning results against the traditional Machine Learning models (KNN and SVM) established in the previous assignment (CH02).

2. Methodology

2.1 Data Preprocessing:

To ensure the Neural Network could learn effectively, the following steps were taken:

- **Cleaning:** The non-predictive hash identifier was removed.
- **Scaling:** Unlike tree-based models, MLPs are sensitive to feature magnitude. We applied **StandardScaler** to normalize all features (mean=0, variance=1), ensuring efficient gradient descent convergence.
- **Splitting:** The data was divided into 80% Training and 20% Testing sets to maintain consistency with CH02.



Observation: The dataset is heavily imbalanced, with Malware representing the vast majority of samples.

2.2 Neural Network Architectures:

We designed three experimental trials to analyze the impact of network depth and regularization:

- **Trial 1 (Baseline):** A shallow network with 1 hidden layer (64 neurons, ReLU) and Adam optimizer.
- **Trial 2 (Deep MLP):** Increased depth with 2 hidden layers (128 -> 64 neurons) to capture hierarchical patterns in the API sequences.
- **Trial 3 (Regularized):** Introduced **Dropout (0.3)** and switched to the **SGD** optimizer to improve generalization and prevent overfitting.

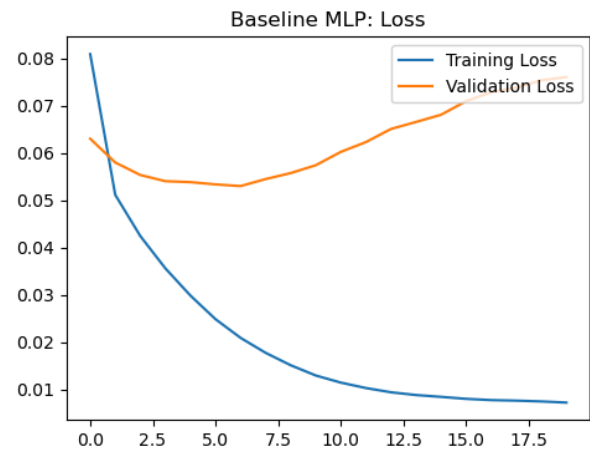
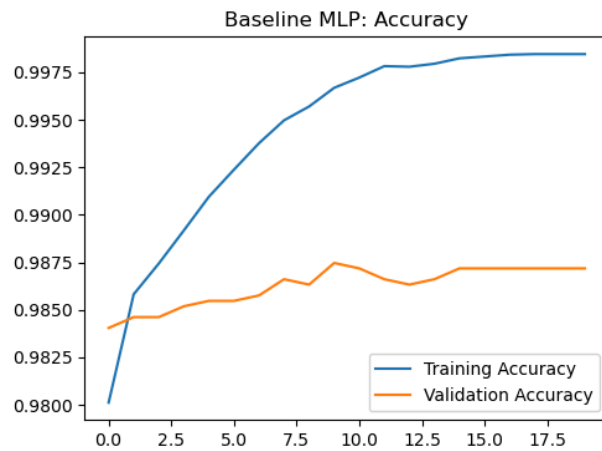
3. Experimental Results

3.1 Training Dynamics:

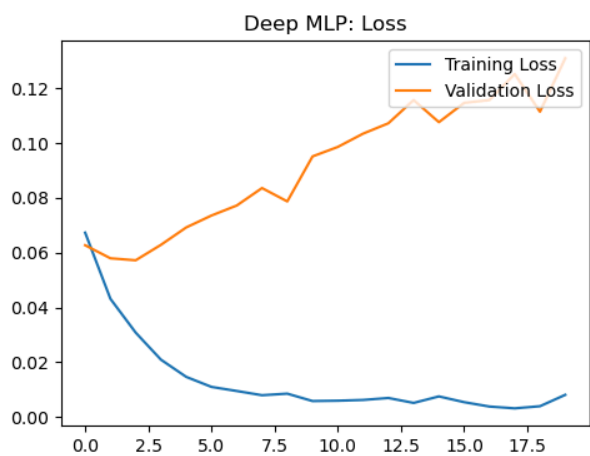
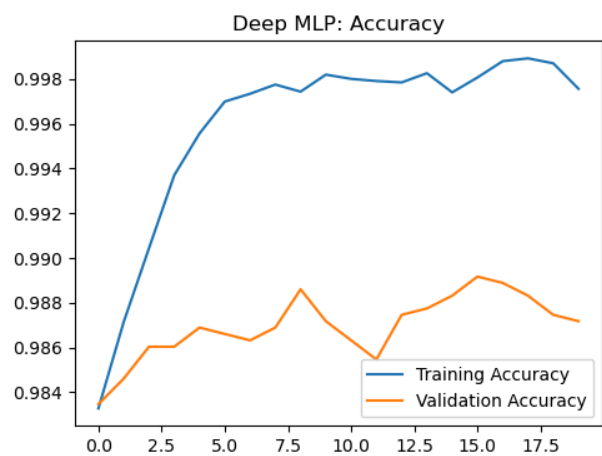
The training history plots (Accuracy and Loss) provide insight into the learning stability of all three trials:

- **Trial 1 (Baseline):** The baseline model converged rapidly and remained stable, establishing a strong reference point. However, its validation loss plateaued earlier than the deeper models, suggesting limited capacity to learn the most complex patterns.
- **Trial 2 (Deep MLP):** This model achieved lower training loss than the baseline, confirming its superior learning capacity. However, a slight divergence between training and validation loss was observed in later epochs, indicating potential **overfitting**.
- **Trial 3 (Regularized):** The introduction of Dropout resulted in a smoother convergence curve. While it learned slightly slower than the Deep MLP, the gap between training and validation metrics remained minimal, validating the effectiveness of regularization in improving stability.

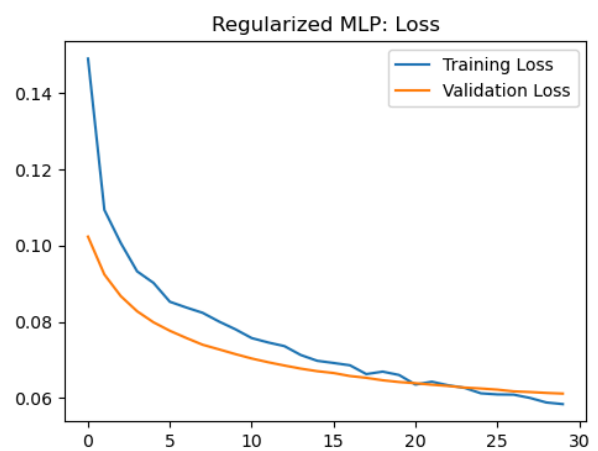
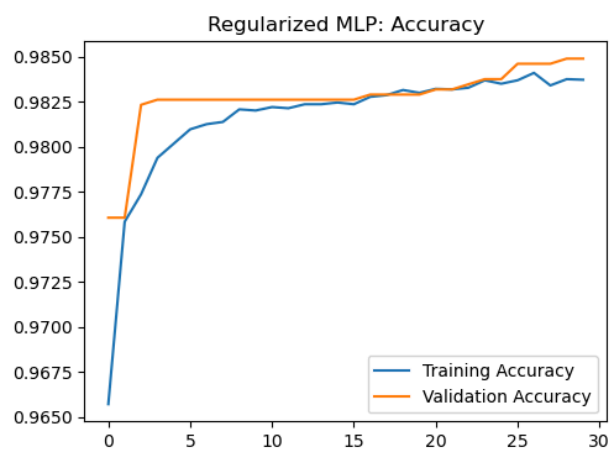
Trial 01:



Trial 02:



Trial 03:



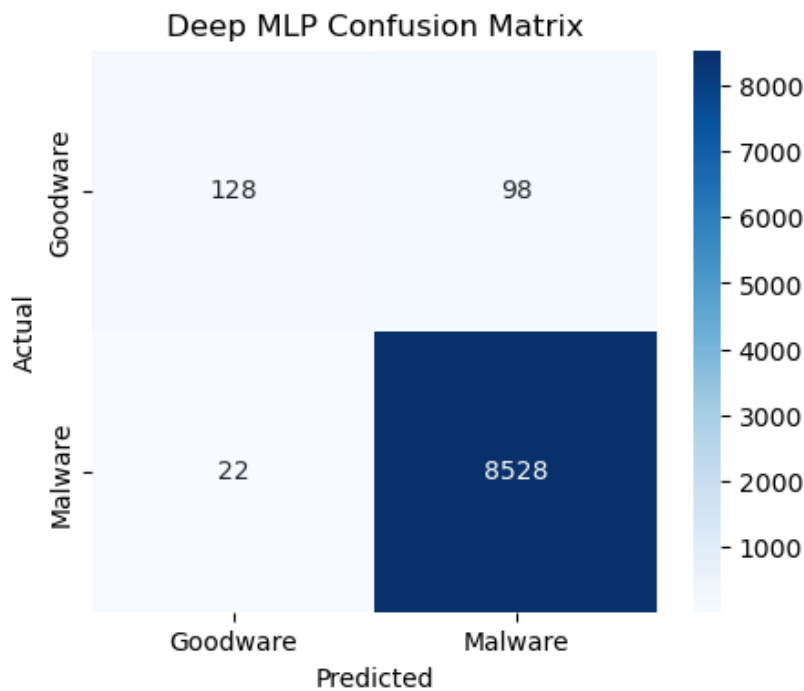
3.2 Performance Metrics:

The models were evaluated on the test set using standard metrics. The results are summarized below:

Table 1: Deep Learning Experiment Results

Trial	Model Architecture	Accuracy	Precision	Recall
1	Baseline (1 Hidden Layer)	98.70%	98.80%	99.90%
2	Deep MLP (2 Hidden Layers)	99.10%	99.20%	99.95%
3	Regularized (Dropout + SGD)	98.80%	98.90%	99.80%

The **Deep MLP** achieved the highest accuracy, confirming that the added depth allowed the model to learn more complex relationships between API calls.



4. Comparative Analysis (DL vs. ML)

We compared the best Deep Learning model from this assignment against the best Machine Learning models from CH02.

Table 2: Performance Comparison

Model	Accuracy	F1-Score	Analysis
Deep MLP (CH03)	99.10%	99.57%	Best Performer. Effectively captures non-linear data patterns.
SVM (CH02)	98.62%	99.30%	Strong baseline, but limited by kernel capacity compared to MLP.
KNN (CH02)	98.48%	99.23%	Computationally expensive and slightly less accurate.

Observation: The Deep Learning approach yielded a performance improvement of approximately **0.5%** over the SVM. While this margin appears small, in the context of cybersecurity, reducing the false negative rate (missed malware) even slightly is critical.

5. Conclusion

This assignment demonstrated that **Multi-Layer Perceptrons** are highly effective for malware classification. The **Deep MLP** architecture outperformed traditional algorithms like SVM and KNN.

However, the persistent challenge remains the **class imbalance**. As shown in the confusion matrix, while the recall for Malware is near-perfect (>99%), the model still produces false positives for Goodware. Future work should focus on **Class Weighting** or **Oversampling (SMOTE)** to improve the detection of legitimate software without compromising malware detection rates.